

Visualizing the chain rule and product rule

The most important chapter for machine learning.

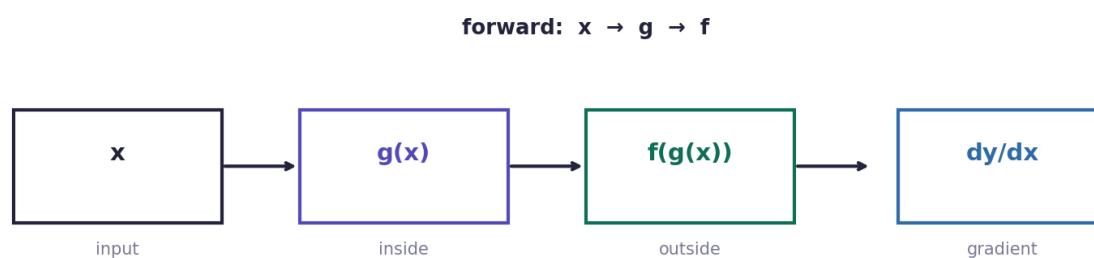
THE IDEA

Two rules let you differentiate combinations of functions: the **product rule** for things multiplied together, and the **chain rule** for functions nested inside functions.

Rule	Formula	Say it as
Product	$d/dx [f \cdot g] = f'g + fg'$	"first prime second, plus first second prime"
Chain	$d/dx [f(g(x))] = f'(g(x)) \cdot g'(x)$	"derivative of the outside, times derivative of the inside"

THE CHAIN RULE, IN PICTURES

A nested function is a pipeline: x goes in, g acts on it, then f acts on that result. To find how the output responds to a nudge in x , you multiply the sensitivities along the pipeline.



backward (chain rule): multiply the derivatives back along the path

WORKED EXAMPLES

CHAPTER 4 — WORKED EXAMPLES

the two rules that make backpropagation possible

PRODUCT RULE

$$d/dx [f \cdot g] = f' \cdot g + f \cdot g'$$

"first prime second, plus first second prime"

example $y = x^2 \cdot \sin(x)$

$$f = x^2 \rightarrow f' = 2x$$

$$g = \sin x \rightarrow g' = \cos x$$

$$dy/dx = 2x \cdot \sin(x) + x^2 \cdot \cos(x)$$

CHAIN RULE

$$d/dx [f(g(x))] = f'(g(x)) \cdot g'(x)$$

"derivative of the OUTSIDE, times derivative of the INSIDE"

example 1 $y = (3x + 1)^5$

$$\text{inside } g = 3x + 1 \rightarrow g' = 3$$

$$\text{outside } f = ()^5 \rightarrow f' = 5()^4$$

$$dy/dx = 5(3x + 1)^4 \cdot 3 \\ = 15(3x + 1)^4$$

example 2 $y = \sin(x^2)$

$$\text{inside } g = x^2 \rightarrow g' = 2x$$

$$\text{outside } f = \sin \rightarrow f' = \cos$$

$$dy/dx = \cos(x^2) \cdot 2x$$

THIS IS BACKPROPAGATION. a neural network is functions inside functions — the chain rule is how the gradient travels back.

Why it matters for ML — this is the chapter that counts. A neural network is a deep nest of functions: layer inside layer inside layer. **Backpropagation is the chain rule** applied repeatedly, multiplying derivatives backward through the network. When gradients "vanish" or "explode", it is because those multiplied factors shrink toward zero or grow without bound.

FORMULAS FROM THIS CHAPTER

Product rule

$$d/dx [f \cdot g] = f'g + fg'$$

Chain rule

$$d/dx [f(g(x))] = f'(g(x)) \cdot g'(x)$$

In words

derivative of outside $\times$ derivative of inside

ML

backpropagation = repeated chain rule

TEST — Chapter 4

1. Differentiate $y = (2x + 5)^3$. $\rightarrow 3(2x+5)^2 \cdot 2 = 6(2x+5)^2$
2. Differentiate $y = x^3 \cdot \cos(x)$. $\rightarrow 3x^2 \cdot \cos(x) - x^3 \cdot \sin(x)$
3. Differentiate $y = \sin(4x^2)$. $\rightarrow \cos(4x^2) \cdot 8x$